

KAI YUE

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Technical Lead

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PROFILE

ML researcher turned AI product engineer and technical lead, specializing in agent workflows, product architecture, and engineering delivery. At Jobdream, translate merchant operations into agent systems and oversee implementation quality. Independently building a game creation and publishing platform, and previously built an investment-education subscription business reaching approximately 300–400 paying members and an estimated \$9,000–\$12,000 in peak monthly recurring revenue. Open to co-founder conversations or joining an early-stage team building agentic products.

BUSINESS & ENGINEERING EXPERIENCE

Technical Lead — Agent Architecture & Engineering Delivery 2025–Present
Jobdream — Private AI education and applied agent systems company, founded in 2020

- Translate B2B merchant operating requirements into modular agent workflows, defining responsibilities for input parsing, low-stock monitoring, and scheduled freshness checks.
- Test merchant workflows hands-on with the team; use findings to refine system behavior, improve usability for nontechnical owners, and assess readiness for delivery.
- Provide technical oversight for outsourced teams of one to several engineers; evaluate vendors and delivery scope, review architecture and code, and make technical acceptance decisions.

Agentic Game Platform Selected Project
Game creation and publishing platform

- Build an agentic platform that lets users create and publish games through vibe coding, coordinating code, 2D animation, shaders, and other assets from concept to playable implementation.
- Use Codex throughout a reviewable feature-delivery loop: translate product intent into scoped changes, inspect architecture and diffs, run automated checks, and iterate against live product behavior.
- Use the platform to develop a LOVE-based game in Lua; refine core mechanics and feed development friction back into improvements to the creation pipeline.

Founder — Investment Education Subscription Business 2023–2025
YouTube audience and paid Telegram membership

- Built and operated a \$30/month membership product, reaching approximately 300–400 paying members and an estimated \$9,000–\$12,000 in peak monthly recurring revenue.
- Grew a YouTube audience to a peak of 9,000+ subscribers and converted viewers into members of the paid Telegram community.
- Deliberately wound down active membership as AI tutoring commoditized the service's core explanations, encouraging members to build and test their own investment models using the principles and Python-based methods taught in the community.

TECHNICAL CAPABILITIES

Product & Agent Systems: Modular agent workflows, workflow validation, architecture and code review, vendor evaluation, and technical acceptance.

Product Stack: TypeScript, React, Next.js, PostgreSQL/SQL; Python for agent services and data workflows.

Additional Languages: Lua, C, C++.

RESEARCH EXPERIENCE

NC State University — Machine Learning Research

2021–2025

Doctoral Research

- Designed federated-learning methods addressing data heterogeneity, communication efficiency, and Byzantine attacks.
- Demonstrated limitations of gradient-obfuscation defenses through image-reconstruction attacks.

UCLA — VCLA Summer Research Intern

Jul.–Sep. 2019

Center for Vision, Cognition, Learning, and Autonomy

- Improved a ROS-based glove interface for real-time hand-pose tracking and interaction in MuJoCo.
- Developed autoencoder-based hand-pose reconstruction in follow-up collaboration, informing later USENIX Security 2023 research.

USTC — Undergraduate Researcher

Aug. 2018–Jan. 2020

MOE–Microsoft Key Laboratory of Multimedia Computing and Communication (MCC)

- Led reference-frame generation in a four-person team; achieved 28% BD-rate savings and second place in WBVC 2018.
- Applied insights from video coding to later predictive-coding research for FL (IEEE JSTSP 2022).

EDUCATION

Ph.D., Electrical Engineering — NC State University

2021–2025

Successfully defended in summer 2025.

B.E., Electronic Information Engineering — USTC

2016–2020

Honorable Title of Excellent Undergraduate Students 2020.

SELECTED PUBLICATIONS

ICML 2025: NTK-DFL: Enhancing Decentralized Federated Learning in Heterogeneous Settings via Neural Tangent Kernel. G. Thompson, **K. Yue**, C.-W. Wong, H. Dai.

USENIX Security 2023: Gradient Obfuscation Gives a False Sense of Security in Federated Learning. **K. Yue**, R. Jin, C.-W. Wong, D. Baron, H. Dai.

ICML 2022: Neural Tangent Kernel Empowered Federated Learning. **K. Yue**, R. Jin, R. Pilgrim, C.-W. Wong, D. Baron, H. Dai.

IEEE JSTSP 2022: Communication-Efficient Federated Learning via Predictive Coding. **K. Yue**, R. Jin, C.-W. Wong, H. Dai. Vol. 16, no. 3, pp. 369–380.

Full publication list on Google Scholar.

SELECTED RECOGNITION

NC State Graduate Award — recurring annual recipient

Through 2025

IYLFs Competition — Third Place

2019

Team-built AI tutoring platform

Tang Lixin Scholarship

2018

China National Scholarship — top 1%

2018